

Assignment 2: Power Electronic Devices and Gate Drive

[Make suitable assumptions, if required]

- Fig. 1 shows a drive circuit for an IGBT. A simplified gate to source voltage V_{GS} is also shown, ignoring the miller capacitance. Assume Q_1 , Q_2 and D_1 to be ideal. Assume first order settling time to be 5τ (τ = time constant). The duty ratio is 0.5.

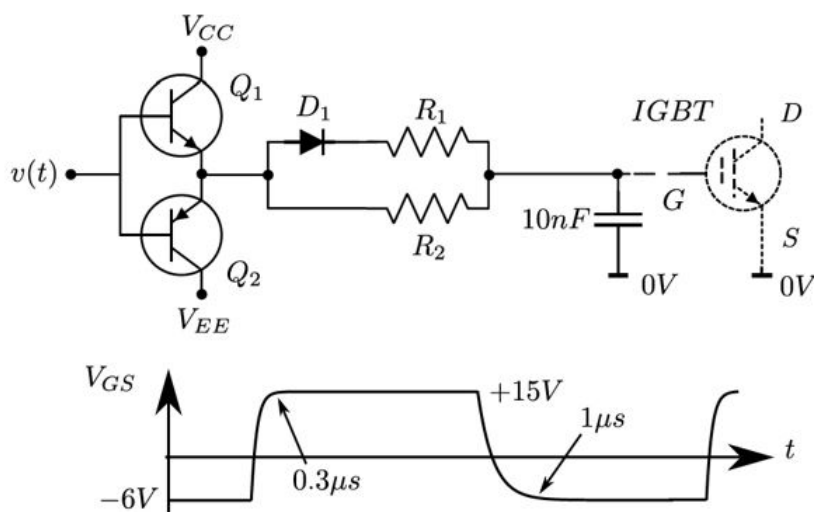


Fig. 1: An IGBT gate drive circuit

- Evaluate V_{CC} and V_{EE} .
 - Compute R_1 and R_2 .
 - Compute the peak current through Q_1 and Q_2 .
 - From the drive current/voltage transients, state if a switching frequency of 20 kHz is satisfactory.
 - Evaluate the average power dissipation in R_1 and the average power dissipation in R_2 at a switching frequency of 20 kHz .
- Fig. 2 shows an SCR, its schematic model, and two periodic operational waveforms. The first operational waveform is a half sinusoid with a peak current of 100 A . The second operational waveform is a pulse width modulated waveform with a peak of 100 A and a duty ratio of 50% . The power dissipation in the device is respectively 56.8 W and 100 W under the two operating conditions.
 - Evaluate the average current and rms current for the two operational waveforms.
 - Evaluate model parameters V and r_d of the SCR.

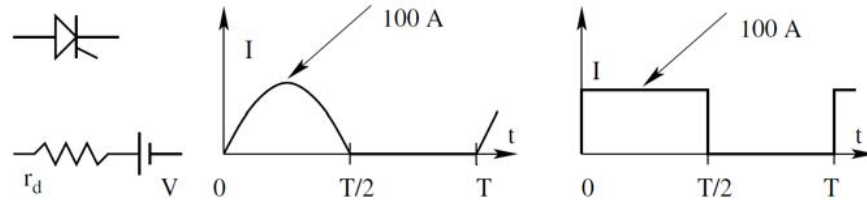


Fig. 2: An SCR, schematic model and two operational waveforms

3. Fig. 3 shows a composite switch consisting of a power MOSFET channel switch in parallel with the body diode. The diode on-state voltage is 1 V. The channel on-state resistance $r_{ds(on)}$ is $50\text{ m}\Omega$. The current i through the switch is a periodic sinusoidal current with peak value of 40 A. Assuming the MOSFET channel to be ON only during the positive half cycle of the current,

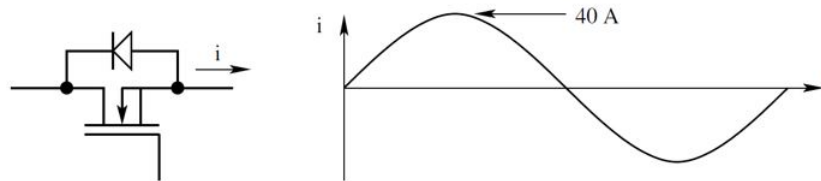


Fig. 3: A composite switch and its periodic current

- A. Sketch the periodic current through the MOSFET channel.
 - B. Sketch the periodic current through the body diode.
 - C. Evaluate the average power loss in the MOSFET channel.
 - D. Evaluate the average power loss in the body diode.
4. The switching voltage and current waveforms of a power switching device with a reactive load is shown in Fig. 4.

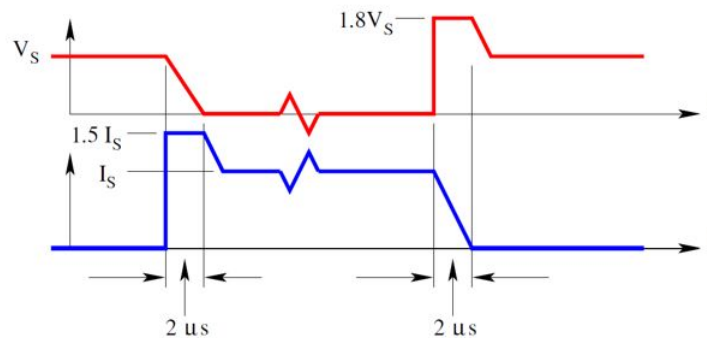


Fig. 4: Switching waveforms of a power switching device

The voltage V_S and current I_S are respectively 200 V and 50 A. The device exhibits overcurrent during turn-on and overvoltage during turn-off as shown in Fig. 4. The switching frequency is 10 kHz.

- A. Sketch the device power dissipation as a function of time.
- B. Evaluate the peak power dissipation during the turn-on transition and the turn-off transition.
- C. Evaluate the turn-on energy loss, turn-off energy loss and the average switching power loss in the switching device.